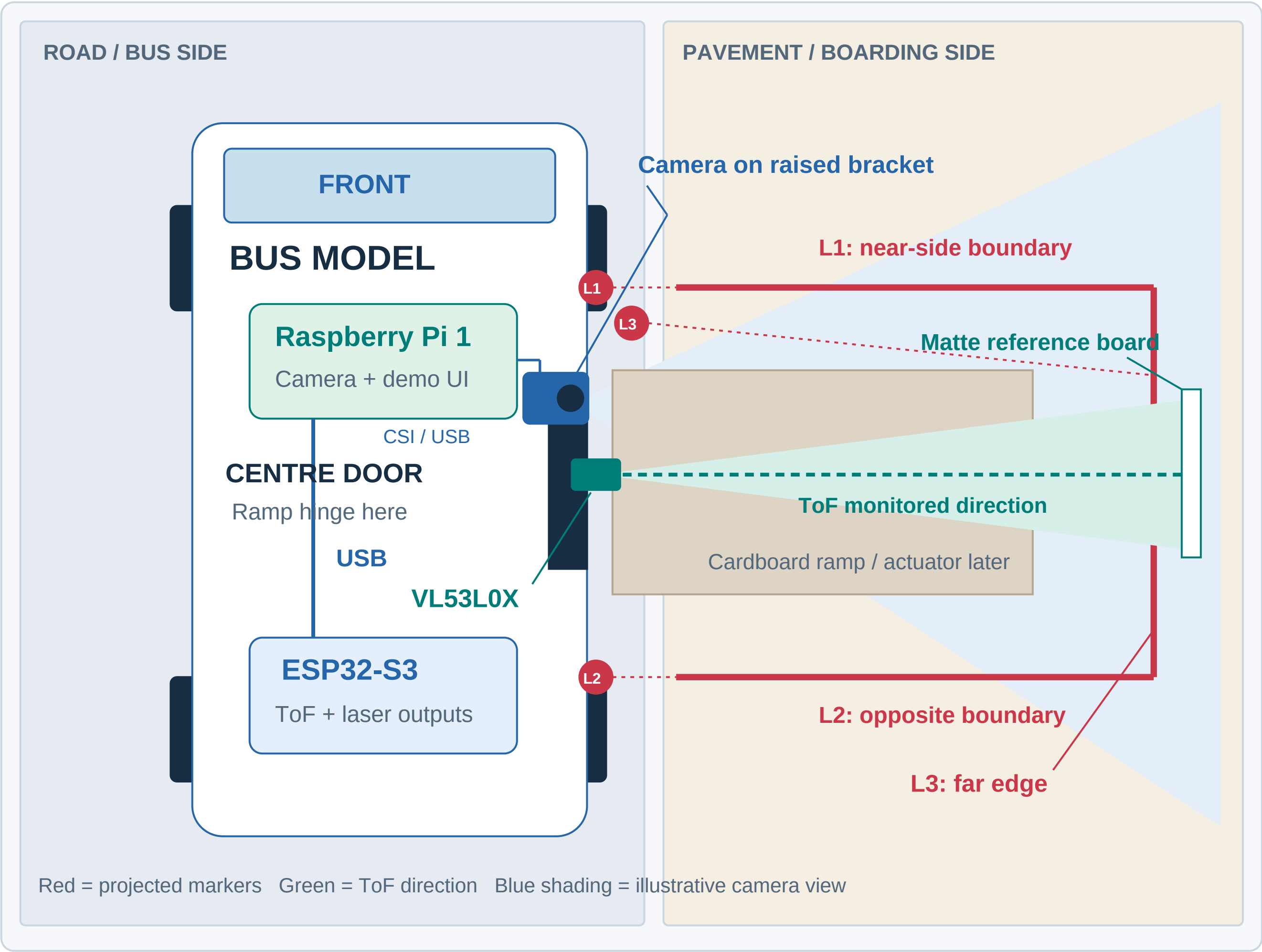


Where everything goes

Top view: bus centre door faces the pavement. A removable roof exposes the Pi and ESP32 for demonstration.



1 Camera and Raspberry Pi

Mount the camera above the centre door, tilted outward to include the waiting area and the whole ramp model. Keep the Pi inside the bus or on the base beside it.

2 Three independent lasers

Mount above the door, aimed down. Your confirmed line modules form a U: L1 + L2 = sides; L3 = the far edge. Rotate each module to align its line.

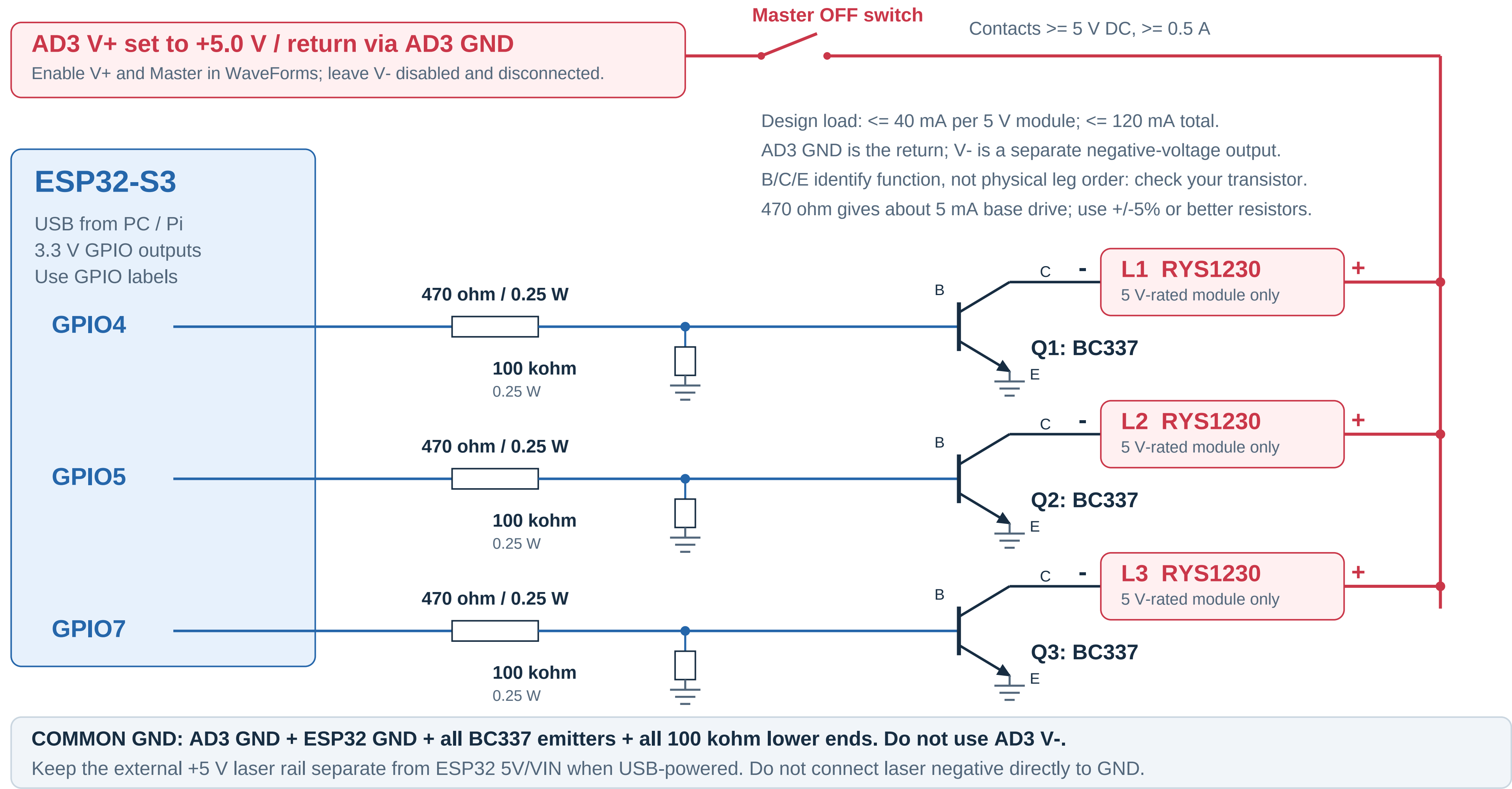
3 ToF detects a test obstacle

Mount beside the door, above the ramp surface, facing a matte reference board. Move a test block into the measured path. The background gives a valid clear reading.

One VL53L0X cannot monitor the full area. A clear beam does not prove a clear zone.

Third laser: repeat the BC337 driver

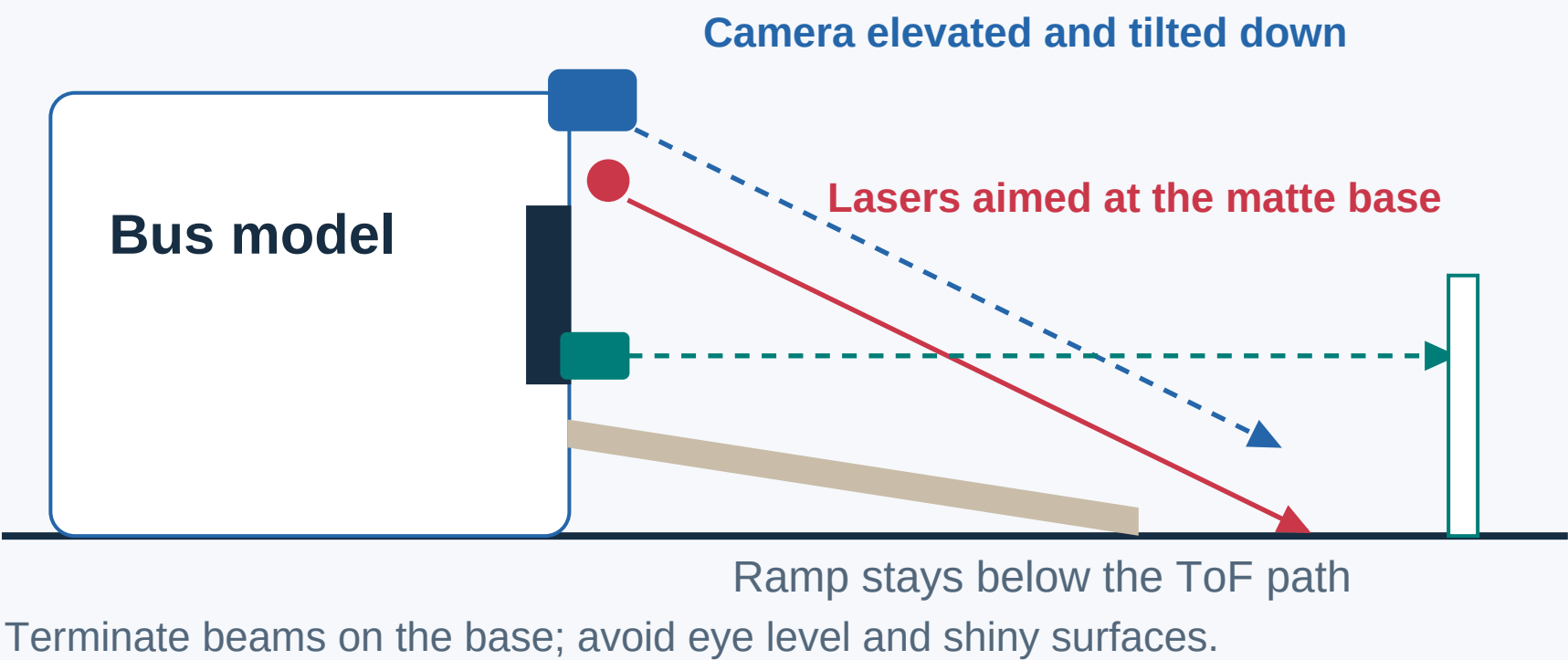
GPIO4 = L1 | GPIO5 = L2 | GPIO7 = L3 | One transistor and two resistors per module



Assemble and demonstrate in stages

Use the same physical layout now; add the request button, limit switches and actuator when purchased.

SIDE VIEW / AIMING



CONNECTION MAP

ToF VIN / GND	ESP32 3V3 / GND
ToF SDA / SCL	GPIO8 / GPIO9
Laser controls L1 / L2 / L3	GPIO4 / GPIO5 / GPIO7
ESP32 USB	Raspberry Pi 1 USB (data + ESP32 power)
Camera	Pi 1 CSI ribbon OR USB, matching camera type
Raspberry Pi power	Its own model-appropriate power supply
Optional Raspberry Pi 2	Request screen; same LAN as Pi 1

BUILD NOW

1. Power off before wiring; AD3 GND goes to ESP32 GND.
2. Secure the bus, camera, ToF and all laser brackets.
3. Check ToF at known distances in the final mounting.
4. Three-laser firmware is now uploaded to the ESP32.
5. Connect Pi 1 to ESP32 USB; launch pi_demo.py.
6. Aim the lasers, then test ON / OFF from the control screen.
7. Set the reference with only the matte backstop in view.

Verified: all three lasers lit, then timed out to OFF.
Pi/camera deployment and final mounting checks remain.

ADD LATER / KEEP LABELLED AS SIMULATED

Physical request button: proposed GPIO6 to GND.
Retracted / deployed switches: proposed GPIO10 / GPIO11.
Actuator: select mechanism, driver and supply before wiring.

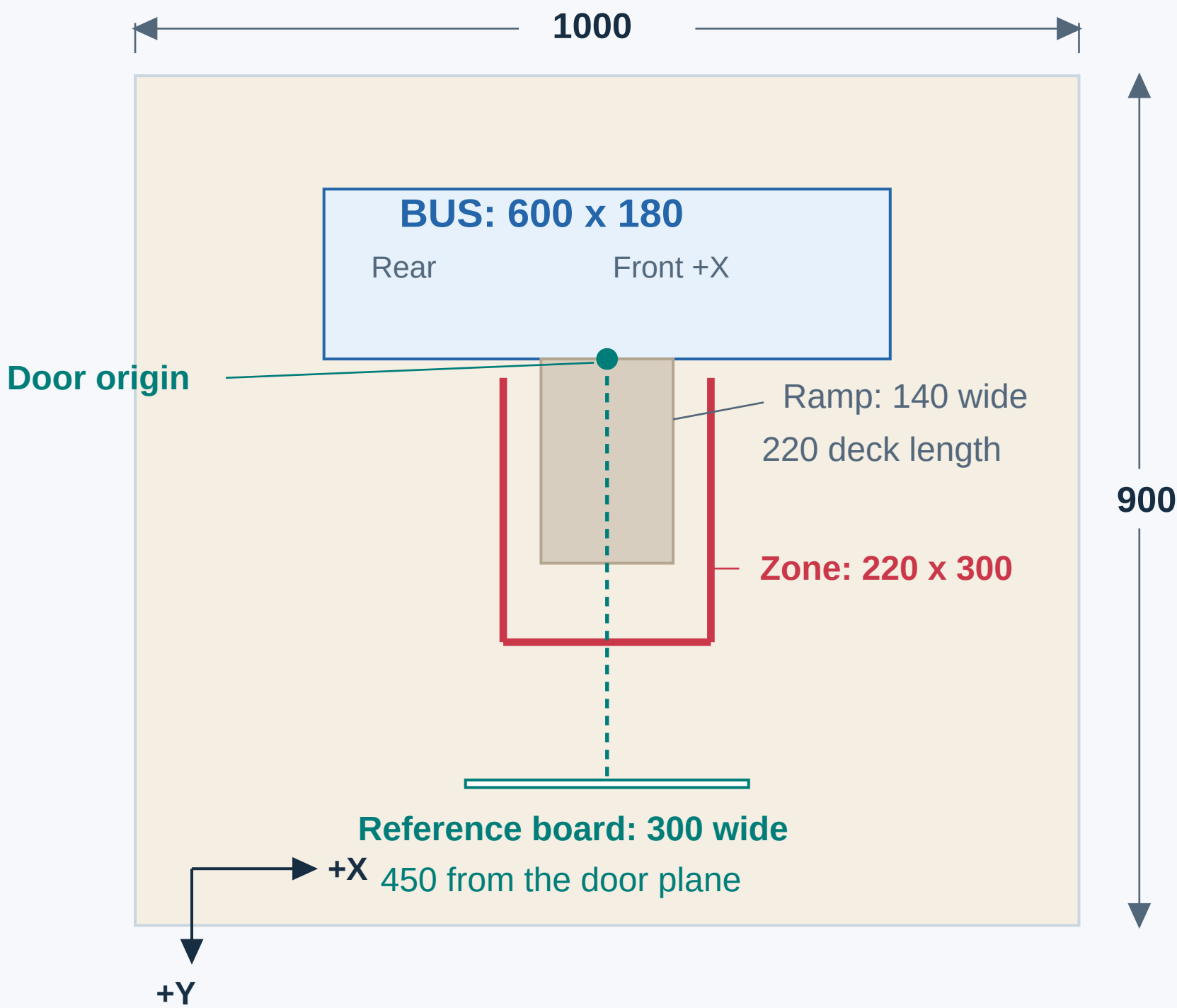
Demo flow: request -> mark zone -> check ToF -> on-screen result.
Use a matte backstop beyond the obstacle test area.
A nearby valid return = blocked; stale/invalid return = unknown.
Only a valid expected backstop return can mean beam clear.

Camera preview is not object detection; ML needs a tested model.

Build dimensions and baseboard layout

All dimensions are in millimetres. These are proposed starting dimensions, not mandatory or scale-certified specifications.

TOP VIEW / DIMENSIONED REFERENCE



Origin on baseboard: 500 from its left edge,
300 from its rear edge; directly below the door centre.
Bus footprint: X=200..800, Y=120..300 on the board.
Z=0 is the top surface of the baseboard.

CUT / BUILD REFERENCE

Baseboard	1000 x 900; thickness 9-12
Bus envelope	600 L x 180 W x 220 H
Centre-door opening	160 W x 160 H
Floor / ramp hinge	40 above the baseboard
Lightweight ramp	220 L x 140 W; thickness 3-5
Laser-marked zone	220 W x 300 outward from bus
White reference board	300 W x 280 H; vertical
Test obstacle	About 150 W x 180 H; stable base

Ramp side clearance: 40 on each side inside the zone.
A 220 deck falling 40 has about 216 horizontal reach.
Bus height 220 includes its elevation above the baseboard.
Choose a lightweight ramp before selecting servo torque.
Dimensions are nominal; allow for actual material thickness.

Mounting coordinates and aiming

Coordinates are relative to the door centre at baseboard level. Z measures the optical centre/aperture, not the bracket.

X: along bus, positive toward front | Y: outward toward boarding area | Z: upward from baseboard

Component	X	Y	Z	Placement / orientation
Camera lens	0	-60	400	Above the roof; aim outward and down
L1 aperture	-110	0	260	Side boundary at X=-110
L2 aperture	+110	0	260	Side boundary at X=+110
L3 aperture	0	+20	280	Far boundary at Y=300
ToF optical window	0	0	140	Level; face straight outward (+Y)
Reference-board centre	0	450	140	Face ToF; board bottom at Z=0
Pi / ESP32	-	-	-	Inside bus; removable roof, USB access, ventilation

AIM THE THREE LINES

L1 / L2: start about 60 degrees below horizontal.
Rotate optics so lines run parallel to the ramp.
Target X=-110 and +110; extend Y about 20..300.

L3: start about 45 degrees below horizontal.
Rotate its line across the ramp at Y=300,
spanning X=-110..+110 to join the two side lines.

Fan angle sets line length. Use adjustable brackets.

CAMERA VIEW AND ToF CHECK

Camera: start about 60 degrees below horizontal,
looking roughly 200 beyond the door. Adjust by preview.

ToF: level at height 140; reference board 450 away.

Nominal 25-degree FoV gives about 200 viewing width
at 450 distance (geometric estimate, not guaranteed coverage).

Use the tall test block; low objects may be missed.

Recheck distances and recalibrate after final mounting.

FoV reference: ST VL53L0X datasheet, performance conditions.